



Notes from the CF Workshop '26 in Bonn

Sadie Bartholomew
CMS Meeting | Friday 2nd October 2026

1) The CF Workshop '26, overall

An overview of the workshop and our involvement



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CF Workshop '26: hybrid meeting at ECMWF Bonn site

Screenshot of <https://climate.copernicus.eu/cf-conventions-community-workshop-2026>



PROGRAMME OF
THE EUROPEAN UNION



IMPLEMENTED BY



21-24 September 2026 • Bonn, Germany • Hybrid Event

Bringing together the global CF community to advance standards, collaboration, and interoperability in environmental data.

- David Hassell and myself on the organising team (~10 people, international mix)
- Both attended in person last week though David had to leave midway for FUTURA meeting in Milan
- Others attending at least parts of the workshop online e.g. Jonny



CF Workshop '26: themes and agenda

- Full details including link to Google Drive with all e.g. slides content on the official meeting page on the CF Workshop shows agenda, etc:
<https://cfconventions.org/Meetings/2026-Workshop.html>
- As lifted from the page above ('Overview'):
 - *"The 2026 CF Conventions Community Workshop is a scientific and technical workshop for the international community working with Climate and Forecast (CF) metadata conventions, environmental data standards, and interoperable data systems.*
 - *The workshop is designed to bring together researchers, developers, data engineers, infrastructure specialists, CF contributors, and users of CF-compliant datasets. The meeting will combine plenary sessions, invited and contributed talks, thematic discussions, breakout sessions, and hands-on hackathons."*
- As lifted from the page above ('Objectives'):
 - *"The workshop aims to support collaboration across the CF community, contribute to the evolution and adoption of CF metadata standards, and make practical progress on CF-compliant data and tools.*
 - ***Themes highlighted for the 2026 workshop include: GRIB to CF, Applying CF to different data formats, Encoding uncertainty metadata in CF.*** Additional topics and community suggestions are welcome..."



Some points of note and outcomes

- Lots of great discussion and some progress. The key now is to keep some momentum going - often it can be lost completely after a workshop due to everyone being busy with their day jobs / other work. I highlight one piece of work at the end of this talk
- Stronger engagement by, and collaboration with, ECMWF
- The WMO announced via Jeremy Tandy that there is a very strong chance they will be happy to host the next in-person CF Workshop (2028)!
- (Tangential) Karl Taylor, one of the key is about to retire - we owe him many thanks



2) What will be in the next release, CF 1.14

Talk for the plenary I gave, on behalf of the CF Conventions community



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A (very) brief history of CF releases

Where are we at? What's our version strategy?

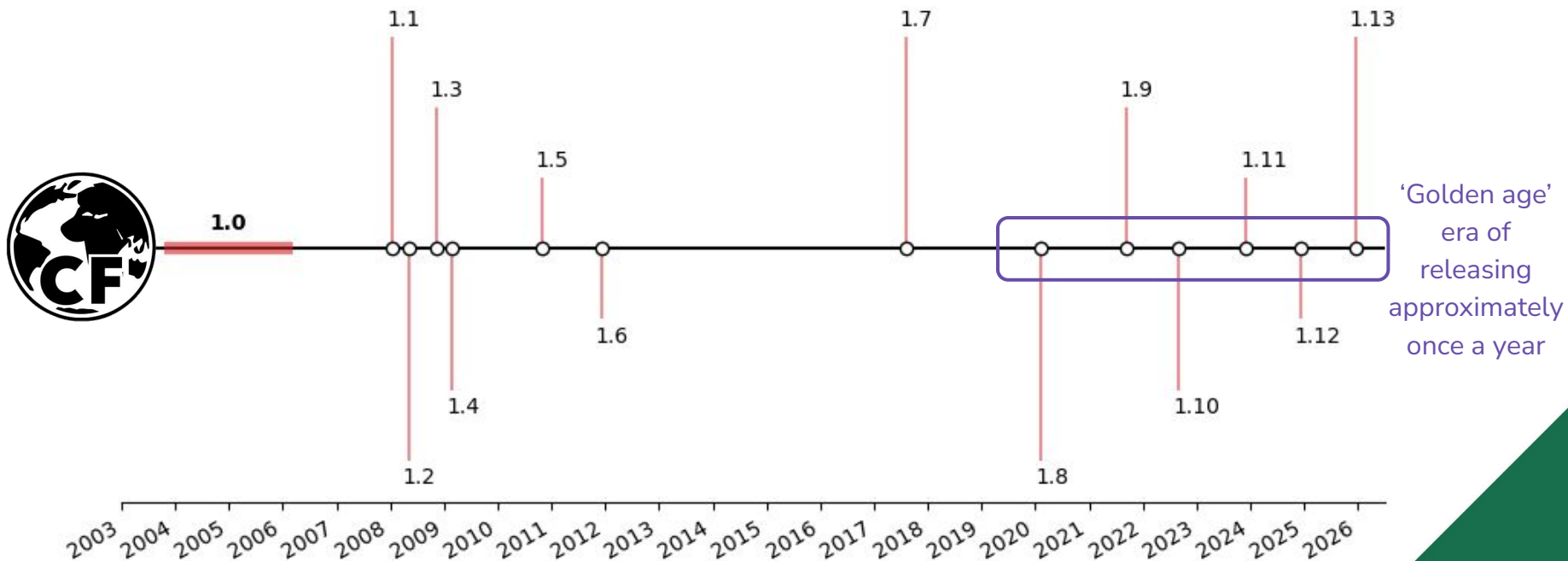


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A timeline of CF (canonical document) versioning*

CF Conventions versions by date



*Note **the controlled vocabularies have their own independent versioning**, mostly since the standard names are released much more often than the Conventions document itself (with current Standard Name Table version v94, released early June 2026)



CF versioning strategy

- **As of 2020**, releases have been every year (quite erratic pattern before then as per timeline of previous slide!)
- **As of 2023**: yearly release date scheduled a *short time after* (previously was *just before*) the annual workshop, to allow items decided on at the meeting to make it more rapidly into the published conventions” on a date agreed during the workshop
- **The idea**: regular releases whereby we can discuss and aim to progress and ideally conclude/merge unresolved loose ends during each yearly workshop
- For example, last year’s workshop was in mid/late September with the release mid December 2025 - so about 3 months later - aiming for similar this year for 1.14



Latest CF Conventions release: CF v1.14, due out in a few months

Replacing current latest CF v1.13 (Dec. 2025)

So what's changing?



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Definitely in (i.e. already merged into) CF 1.14

New items are listed in the 'Revision History' at the very foot of the (draft version) canonical conventions document, in this case for v1.14 at: https://cfconventions.org/cf-conventions/cf-conventions.html#_version_1_14_draft, or you can consult the GitHub milestone for any (made or upcoming) release, for v1.14: <https://github.com/cf-convention/cf-conventions/milestone/9>

- **Reduced Gaussian grid** support [\[#619\]](#)
 - no existing CF grid mapping can capture the characteristics of the reduced Gaussian grid, so a new convention and parameters are adopted to encode it within CF
- A new parametric vertical coordinate, '**atmosphere hybrid sigma log-pressure coordinate**' [\[#647\]](#)
 - Support for a vertical coordinate type used by the GEM atmospheric general circulation model. See new drafted sub-section under 'Appendix D: Parametric Vertical Coordinates'.
- **Clarify units prohibition** of numerical factors and offsets [\[#620\]](#)
 - CF explicitly imposes a restriction on a subset of the syntax that UDUNITS permits
- Section '**4.3. Vertical (Height or Depth) Coordinate**' clarity improvements [\[#635\]](#)
- **Formatting/consistency and infrastructure** improvements and some "bells and whistles":
 - Consistently reference "CF conventions" not "standard" or just "convention", and use the impersonal form ("we recommend" → "it is recommended") [\[#623\]](#)
 - Consistently represent datetimes and use calendar attribute in examples [\[#641\]](#)
 - Configurable dark mode for HTML view [\[#607\]](#)
 - Updated website e.g. homepage improvements (not technically tagged with v1.14 but new since last year) [e.g. cf-convention/cf-convention.github.io#605]



Some definite inclusions, illustrated

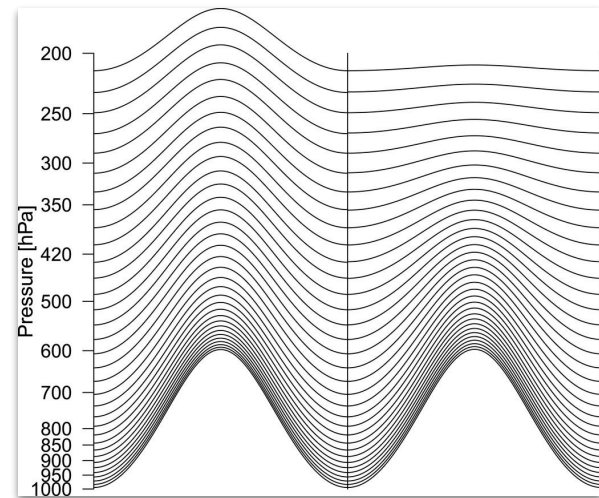
Atmosphere hybrid sigma

log-pressure coordinate

Rectification coefficients control how rapidly the vertical levels are rectified over the mountain.

“Vertical model-level structure showing ζ levels as functions of pressure for values below 200 hPa ...in a typical operational configuration [for an illustrative] sinusoidal mountain” from

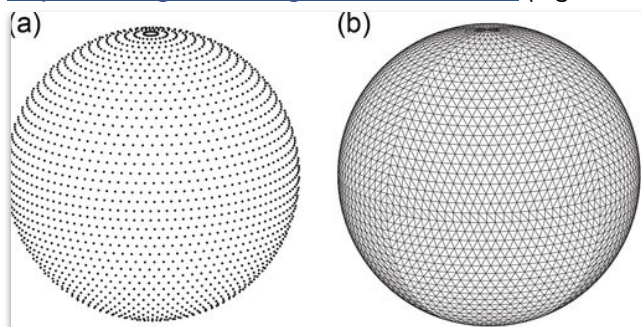
<https://doi.org/10.1175/MWR-D-13-00255.1>:



Reduced Gaussian grid support

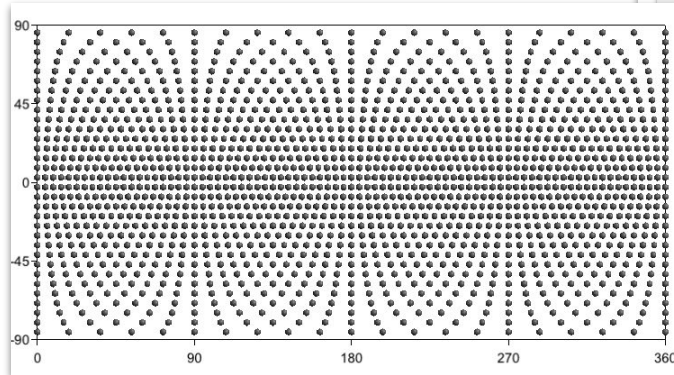
Octahedral Gaussian (O24) grid example from

<https://doi.org/10.5194/gmd-12-651-2019> (Fig. 2, cropped):

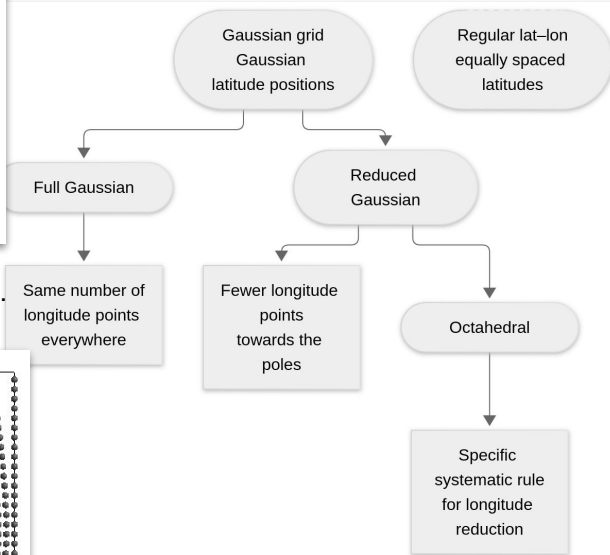


O16 grid example shown in projected lat vs. lon. form.

From: <https://sites.ecmwf.int/docs/atlas/design/grid/>:



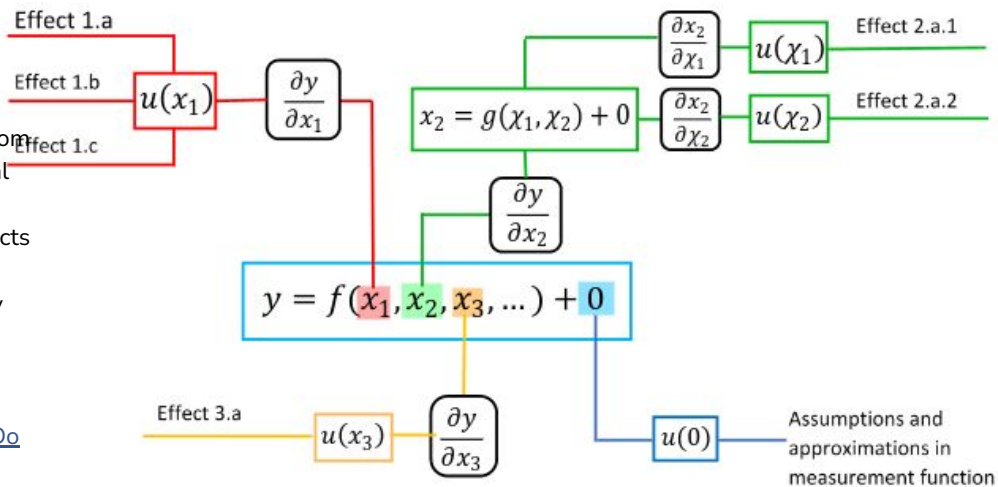
Schematic illustrating Gaussian grids e.g. how the 'reduced' case differs from 'full' case:



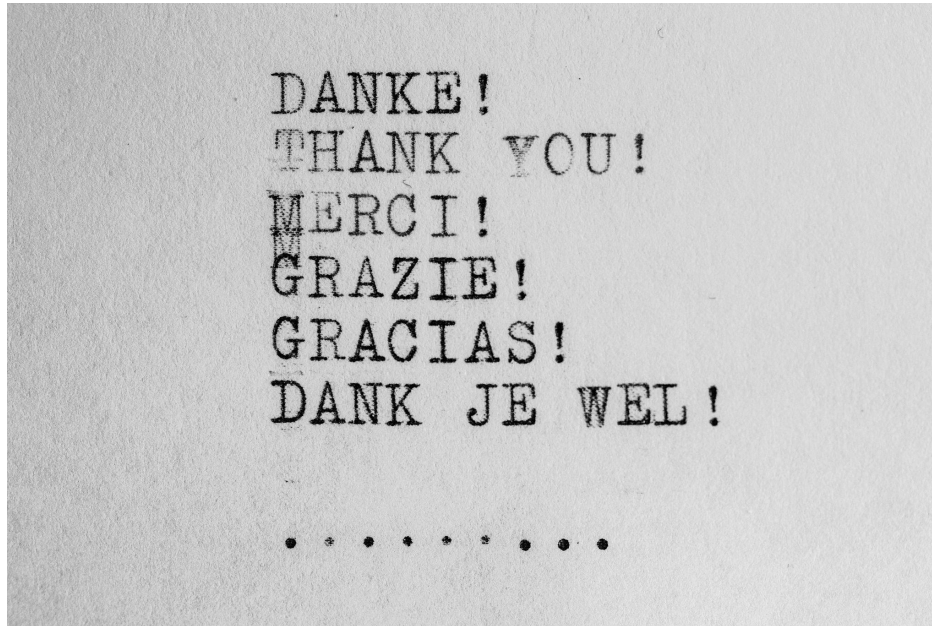
Major work in progress, unlikely to make it to CF 1.14

- Uncertainty metadata [[#660](#)]. As copied verbatim:
 - CF still needs to **formalise support for metadata that describes the uncertainty in the measurement of a quantity** described by a data variable
 - Proposed to do so with the introduction of a *new uncertainty variable*, which is a container for uncertainty information. It may be thought of as a special type of ancillary variable, in that it provides metadata which varies over the sampling domain of a data variable. An uncertainty variable has standardised attributes to describe the nature of the uncertainty, and its data array provides the actual uncertainty values.
 - The probability distribution of the quantity being measured (i.e. the measurand) can be described either not at all, very simply (e.g. just noting that it is Gaussian), or in great detail (e.g. by kurtosis parameter values which vary over the domain, and error-correlation coefficients between arbitrary dimensions)
 - Builds on work by Sam Hunt and Pieter de Vis towards the [Uncertainty Metadata Naming Conventions \(UNC\)](#)

Illustration of the potential complexity of uncertainty quantification via a 'Conceptual Uncertainty Tree Diagram', taken from 'General guidance on a metrological approach to fundamental data records (FDR), thematic data products (TDP) and fiducial reference measurements (FRM) – Uncertainty Analysis Process' by the UK's National Physical Laboratory (NPL)QA4EO Team, at https://qa4eo.org/docs/3_Process_Document.pdf:



Many thanks to everyone who contributed in any way to CF 1.14 (or earlier)! CF is a community effort and we rely community contributions*



*or will contribute still - you have a little time left!



In summary

- Since 2023 we've made **yearly releases a few months after each annual workshop**
- **CF 1.14** is this year's imminent version, replacing Dec 2025 version 1.13
- Changes *definitely* going into 1.14 (already merged ready for release) include:
 - Reduced Gaussian grid support [[#619](#)]
 - A new parametric vertical coordinate, 'atmosphere hybrid sigma log-pressure coordinate' [[#647](#)]
 - Clarify units prohibition of numerical factors and offsets [[#620](#)]
 - Section '4.3. Vertical (Height or Depth) Coordinate' clarity improvements [[#635](#)]
 - Formatting/consistency and infrastructure improvements
- Standardisation of uncertainty metadata [[#660](#)] is a major endeavour in progress but is unlikely to make it into v1.14



3) Adoption of BCP 14 requirement levels in CF

- a) Introduction and status to lead into a breakout session I hosted



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The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

- **BCPs** are stable identifiers for Best Current Practices
- We want to apply these to the full CF document, see [Issue #258](#) (started 2023)
- [BCP 14](#) consists of two related standards:
 - **RFC 2119**: Defines terms like MUST, SHALL, REQUIRED, SHOULD, RECOMMENDED, MAY, and OPTIONAL to establish strict technical requirements
 - **RFC 8174**: Updates RFC 2119 by clarifying that these words only carry their special requirement definitions if written in all-uppercase (e.g. MUST)



Illustrating the concept

- **should** ≠ **SHOULD**, conveying different intention and context:
 - From general conversation: “You **SHOULD** back up any important files regularly.” [if give it BCP intention], vs. “The train **should** arrive on time.” [expectation, not a recommendation], vs. “It’s raining so you **should** bring an umbrella” [advice, not formal intention]
 - From CF: “The conventions **should** be practicable for both producers and users of data.”, “The metadata **should** be both easily readable by humans and easily parsable by programs.”, but “The units attribute **SHOULD** be present for all variables that contain dimensional quantities.”
- Note: one may not use the exact words, but regardless have the same meaning, one goal of the hackathon could be to identify the underlying requirement level, rather than mechanically replacing words. Example: “**It is suggested that** a zero offset be stated explicitly to avoid confusion...” is a RECOMMENDED case without using the word itself.



For the breakout/hackathon session

- **This topic was a breakout session at the 2024 CF Workshop** and good progress was made, see https://docs.google.com/document/d/1VknWozJM3vkUeoJKQHXLmroXPwZtd2EO8_jgGIIID-kw/edit?tab=t.0 (GDoc notes) and <https://github.com/cf-convention/cf-conventions/issues/546> (follow-on Issue for implementation, now closed as “not planned”). Daniel Lee’s summary of status:
 - *We've decided that we want to use BCP 14 unmodified, and that the "EBCP" words are to help us ensure that we're actually saying what we mean. The rendering is still an open issue, although I believe we can address that a bit further down the road.*
 - *“The rendering”: some thought all upper case is “too shouty” so TBC whether to adopt that i.e. RFC 8174 way to indicate RFC 2119*
- **Lars has done excellent work to analyse the document to e.g. provide counts of the BCP keywords** so much work is required for a given one in any conversion
- **BCP adoption can't go in the document piecemeal**, so aim to get a Pull Request fully implementing it prepared to be merged and adopted in time for the release next year (v1.15), ideally
- **First and foremost, we need to agree a strategy for getting BCP in.** Suggestion to get the ball rolling!
 - Identity all statements with the BCP keywords or equivalent wording (Lars’ work helps)
 - Big table of all of these with checklist to tick off thoughts of BCP case or not?
 - In the age of AI tools/agents, could we use these to make a first pass and then have N (2?) humans confirm if they agree - could speed up the process?



3) Adoption of BCP 14 requirement levels in CF

b) My notes from the breakout session itself



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Summary (raw) notes from the breakout session (1/2)

- Are they testable, i.e. from Conformance Document, could that be a way to do a first pass - only “testable” docs. But it isn't clear what is or isn't testable in many cases - needs debate.
- Could we start with the conformance doc - that is definitely a technical doc!
 - Could we even combine the conf. & canonical doc, s.t. any BCP constitutes a formal statement — instead of separate docs → these could be, say, rendered as “admonitions” — coloured boxes.
 - If conf. pulled into main/canon doc, must have a way to extract out so tools can parse without having to wade through the non-conformance content.
 - Conf. doc just has “reqs” and “recs”: could be trivial. Question is: does conf. doc use BCP words → BCP way right now?
- (Some) statements need reviewing for intention → should or “may be” might be written but the intention might have been for something more or less strong.
- Are should + recommend interchangeable — which to use if so?
 - Contrasting views on which to use ('should' or 'recommend') in that case (should could imply “must”)
 - Similarly, do we choose “shall”: over “recommended” etc for synonyms?
- Though BCP 14 defines the terms, ideally can be clear standalone e.g. when being discussed verbally
- WMO docs (WIS2) use BCP — manual, lifted from OGC way to work. They include conformance statements there, using boxes to separate them- MAY is called “a permission” case in those WMO docs
 - WMO have CI verifier - was suggested too in 2024 CF workshop.
- "Small print" / license / legal doc like - will people even read that?
 - → Only use those keywords in a BCP sense! → then don't need upper case but think it is good to.
 - → still have the “headings” in them? → or in a table or similar, as a way to distinguish → can add/convert.
- Would be good to convert also the standard names / vocabs (maybe some of the area types too) descriptions



Summary (raw) notes from the breakout session (2/2)

- **Strategy/plan for going forward:**
 - Idea: convert wholesale & review if anything is wrong (i.e. all to uppercase).
 - Incremental strategy: by chapter? by conf. statements?
 - Convert term by term or by chapter?
 - Working groups/editorial board? Need to make sure changes for v1.15 are included - do these at the end, near the release time.
 - Use git branching — separate branch.
 - Could conf. doc (trivial?) go into version 1.14 instead of 1.15?
 - Could use GitHub approvals for tick/cross on changes.
 - As well as converting, existing docs need a way to ensure changes include correct BCP.
- **Decisions(?):**
 - Conformance docs first.
 - Do use caps/uppercase.
 - Put on issue about a working group with regular meetings (monthly?) - go chapter by chapter or similar→ chunk out in sensible pieces.
- **TODOs**
 - Recruit for WG + pick dates / scope (chapters etc.)
 - Try to summarise hackathon session thoughts/ideas/prog.
 - Ask in issue, broadcast W.G. when feed back on session → & use auto-tools to put in caps, only to downgrade if agree.

